Classifying Voice Commands Using CNN Machine Learning

Benjamin Bellenchia, Brian Patterson, Eric Dutan, Kai Perez

In Collaboration with Dr. Elgala

INTRODUCTION

The goal of this project is to recognize spoken commands such as “yes”, “no”, ”stop”, and ”go” using machine learning.

- Raw audio signals are converted to Mel Spectrogram images
  - Mel Spectrograms are used to detect the change in change in frequency over time
  - We use Mel Spectrograms to compare audio samples and train the model to detect the similarities between the word and audio file given
- Model receives command using microphone post training and determines if the word is a command and which one

Dataset Used

- Used Google Speech Commands Dataset v0.02 containing short spoken word recordings
- Dataset was split into 80% training and 20% testing

MODEL ARCHITECTURE

Support Vector Machine (SVM) Model

- Model created to test differences between CNN based model and other machine learning methods.
- SVM model proved to be highly inaccurate compared to CNN (difference of ~20%).

Voice Recognition

- Data:
  - 25 Audio samples of team saying “UAlbany”.
- Scaled signal between [-1, 1] and standardized 16kHz sampling rate between each recording
- Uses Gaussian Mixture Model (GMM) to detect different users.
  - Works by “soft-clustering” all data presented and the probability it is a specific user.

CONVOLUTIONAL NEURAL NETWORK (CNN) MODEL

- CNN Model is extremely accurate at determining if a word is apart of the commands list.
- Model reaches 95% accuracy at detecting if the spoken word is one of the commands as well as “silence” or “unknown”.
  - Allowing model to determine these is pivotal to increasing accuracy among the commands it does know.
- Noise is added into the audio files during processing to teach the model to separate background noise from commands given.

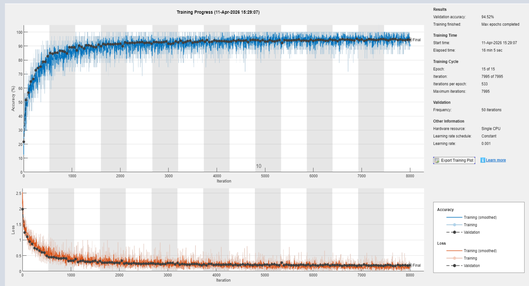

Figure 1: Results after 3rd Convolution

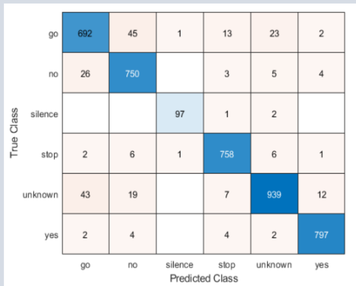

Figure 2: CNN Confusion Matrix after 3rd Convolution

MODEL FLOW CHARTS

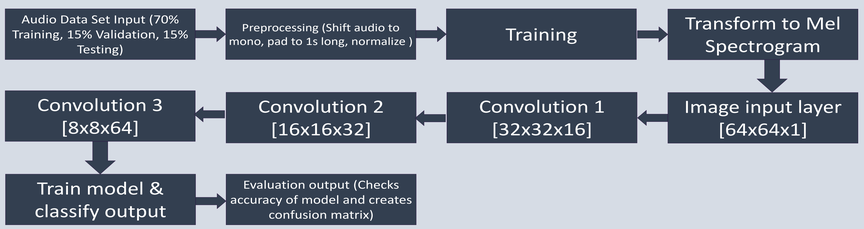

Figure 3: CNN Model Flow Chart

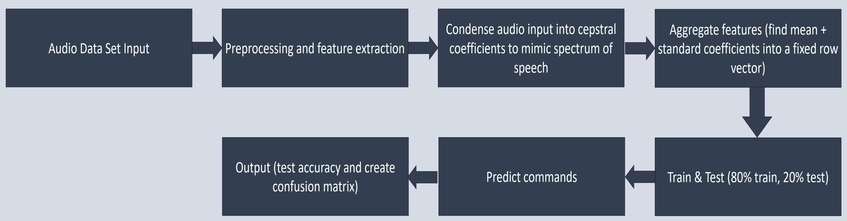

Figure 4: SVM Model Flow Chart

RESULTS

Real Time Prediction

- After training the model we provide an audio file of a command and test if the CNN model can correctly predict which command it was or if it was a command at all.
- After the first Convolution the model is 84% accurate.
- By the second training accuracy improves to ~92%.
- When adding a third layer, noise, and a drop out layer the model is 95% accurate
- The confusion matrixes between each model show each struggle to differentiate “go” and “no”. CNN is 34% and 28% more accurate in each.

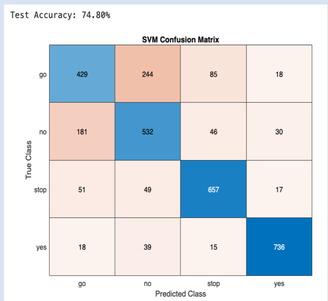

Figure 5: SVM Confusion Matrix

CONCLUSIONS/FINDINGS

- CNN architecture is the best for supporting our projects requirements.
- Added noise, another convolution layer. and allowing model to be “uncertain” increased accuracy by ~4%.
- Voice recognition adds extra depth to the system allowing specific users to use it at a time depending on constraints given to system.